Beyond Semantic Confidence: Relation-Consistent Geometric Re-ranking for 3D Scene Graphs

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Abstract

3D Scene Graph predictors can assign high scores to relation candidates whose predicates are inconsistent with reconstructed ordered-pair geometry. We formulate predicate–geometry compatibility assessment for fixed relation predictions and introduce RelCompat3D, which preserves ordered-pair identity and separates predicate/family semantics, predicate-independent pair-geometry measurements, and the source relation score. Family-specific compatibility heads are trained using linked positive–counterfactual pairs without predictor identity or the source relation score. Transformation averaging enforces proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation. The method re-ranks proximity and vertical relations while keeping support/contact relations in source order. We evaluate exact-label Recall@K jointly with verifier-derived Violation@K across VL-SAT, Open3DSG, and SceneGraphFusion (SGFN) on a shared 3DSSG target. Across the reported budgets, RelCompat3D yields lower or tied Violation point estimates while preserving or improving Recall point estimates, with the largest gains on Open3DSG. Comparisons with matched rank-fusion and nonlinear baselines reveal predictor- and family-dependent trade-offs.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be supported by the reconstructed scene, not only plausible at the category or language level. A graph edge such as `standing on`, `close by`, or `higher than` is useful only if it describes the corresponding ordered subject–object pair in 3D.

We study a failure mode in which relation predictors rank plausible labels that are inconsistent with measured object-pair geometry. A relation can sound plausible for two object categories while being inconsistent with measured distance, projected overlap, or vertical arrangement; contact-heavy relations are assessed separately by the verifier. This mismatch becomes consequential as 3D Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a source relation score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a high-scoring symbolic relation that contradicts the reconstructed scene evidence.

This failure suggests separating predicate semantics T, predicate-independent pair-geometry measurements G, and the source relation score Z. RelCompat3D preserves each candidate's subject–object identity and learns predicate–geometry compatibility from T and G without using Z or predictor identity. Linked positive–counterfactual pairs optimize the ordering of geometrically compatible and incompatible candidates; averaging over known endpoint/predicate transformations enforces proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation. These choices address distinct failure modes: excluding Z precludes direct copying of the predictor score, preserving ordered-pair identity prevents scene-level measurements from substituting for instance evidence, and transformation averaging assigns equal compatibility to equivalent endpoint/predicate representations.

The source relation score enters only during re-ranking. Within proximity and vertical relations, RelCompat3D combines it with compatibility and changes the order within each relation family. It preserves the sequence of relation-family labels in the source ranking and the source order of support/contact relations. Matched fusion baselines and ablations test the effects of excluding the source relation score, preserving ordered-pair identity, and enforcing transformation consistency. Figure 1 summarizes the framework.

We evaluate support/contact, proximity, and vertical-order

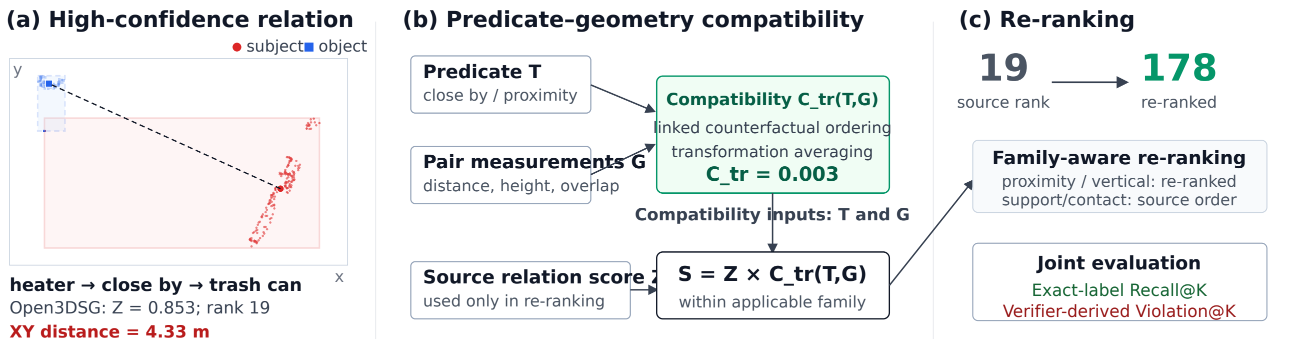

Figure 1: **Failure mechanism and RelCompat3D overview.** A high-scoring Open3DSG relation is inconsistent with the measured geometry of the corresponding ordered pair. RelCompat3D preserves ordered-pair identity, separates predicate semantics T, predicate-independent pair-geometry measurements G, and source relation score Z, and estimates the transformation-consistent compatibility $C^{\mathrm{tr}}(T, G)$ without Z. Family-aware re-ranking moves the shown proximity relation from rank 19 to 178 while preserving the source relation-family sequence and keeping support/contact relations in source order.

relations because their consistency can be assessed from reconstructed ordered-pair geometry. Open3DSG, VL-SAT, and SceneGraphFusion (SGFN) provide complementary open-vocabulary and closed-set predictors (Wang et al. 2023; Koch et al. 2024b; Wu et al. 2021). We jointly report exact-label Recall@K and verifier-derived Violation@K across the three predictors on a shared 3DSSG target.

Our contributions are threefold:

1. We define and measure a mismatch between relation scores and instance-level geometric consistency that exact-label Recall alone does not capture, using identity-preserving evaluation and joint exact-label Recall@K–verifier-derived Violation@K.
2. We learn predicate–geometry compatibility from the corresponding ordered pair without using the source relation score, and enforce exact endpoint/predicate transformations through transformation averaging.
3. We introduce a constrained re-ranking rule that preserves the source relation-family sequence and support/contact order, and evaluate one fitted compatibility layer across three predictors to characterize predictor- and family-dependent gains and failures.

Across $K = 10$–50, no predictor's Recall point estimate decreases, and at $K = 50$ paired scan-cluster intervals support lower Violation without a detectable Recall loss. RelCompat3D re-ranks proximity and vertical relations while preserving support/contact source order; it does not replace the relation generator.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSSG setting introduced a benchmark and evaluation setting for 3D relation prediction (Wald et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multimodal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods already use geometry; RelCompat3D asks the complementary question of whether a fixed high-scoring predicate is geometrically compatible with the corresponding reconstructed ordered pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023); graph systems then connect such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, it becomes important to assess whether plausible text labels are consistent with the reconstructed geometry of the corresponding ordered pair.

Geometry-aware relation evidence. 3D relation prediction and graph reuse already rely on edge features, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Xie, Pagani, and Stricker 2024). Adjacent work on object placement and affordance grounding also uses support/contact patterns and affordance geometry (Huang et al. 2025; Shao et al. 2025). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness integrates visual-geometric witnesses into open-vocabulary 3D scene graph learning and decoding

(Nguyen et al. 2026). Statistical confidence rescoring learns source-specific refinement from semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). PUF introduces training-free uncertainty-aware fusion for online scene-graph association and evidence accumulation (Yang et al. 2026). RelCompat3D instead evaluates fixed relation candidates using predicate–geometry compatibility whose inputs exclude predictor identity and the source relation score.

Several recent systems incorporate geometry or transformation structure directly into relation generation. GEODE conditions predicate generation on updated geometry tokens inside a discrete-diffusion generator (Feng et al. 2026), while RelGraphOV prunes geometrically implausible connections when learning an open-vocabulary scene representation (Chen et al. 2026). Transformation-Aware Decoupling separates predicate reasoning according to how labels transform under yaw-viewpoint changes inside the generator (Sun et al. 2026). In contrast, RelCompat3D leaves each predictor fixed. After prediction, it averages compatibility over applicable joint transformations of the predicate and ordered-pair geometry, while excluding the source relation score from compatibility inputs.

Neau et al. mine spatial, functional, and relation-algebra constraints and use declarative reasoning to refine fixed image-SGG outputs, reporting a Constraint Violation Rate alongside task metrics (Neau et al. 2026). RelCompat3D differs in estimating continuous compatibility from reconstructed 3D geometry, preserving ordered-pair identity, and enforcing applicable endpoint/predicate transformations before joint Recall@K–Violation@K evaluation.

Reliability evaluation. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is geometrically consistent with the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017). RelCompat3D instead associates fixed relation predictions with 3D evidence from the corresponding ordered subject–object pair and reports exact-label Recall@K together with verifier-derived Violation@K. Its compatibility output is a bounded ranking score, not a probability of physical validity.

Method

RelCompat3D assesses fixed relation predictions rather than replacing a 3D Scene Graph generator. It preserves candidate identity, estimates predicate–geometry compatibility without the source relation score, and combines the two only during re-ranking (Figure 1).

Problem Setup

Let a relation predictor produce relation candidates

$$r_i = (\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i, p_i, a_i, Z_i), \tag{1}$$

where s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate, a_i its mapped relation family, and Z_i the source relation score. A context is a scene subgraph over which the predictor ranks candidates; top-K selection is performed independently within each context. The tuple $(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i)$ uniquely identifies the ordered object pair and is retained when associating each prediction with geometry and ground truth. We represent predicate/family semantics as $T_i = (p_i, a_i)$ and associate predicate-independent pair-geometry measurements

$$G_i = G(\mathrm{scan}_i, \mathrm{context}_i, s_i, o_i). \tag{2}$$

The measured scope is

$$\mathcal{A} = \{\text{support/contact},\ \text{proximity},\ \text{vertical order}\}, \tag{3}$$

whose consistency can be assessed from reconstructed ordered-pair geometry. Candidates outside these families are excluded from the measured violation claim. The re-ranking scope is narrower: $\mathcal{A}_{\mathrm{tr}} = \{\text{proximity}, \text{vertical order}\}$, for which the defined endpoint/predicate transformations are exact. Support/contact is evaluated but kept in the source ranking because no single endpoint transformation preserves the semantics of every predicate in this family.

Our objective is to estimate a bounded compatibility score

$$C_i = \sigma(h_{a_i}(\Phi(T_i, G_i))) \in [0, 1], \tag{4}$$

then average over the valid family-specific endpoint/predicate transformations to obtain the transformation-consistent score C_i^{tr} and form a final ranking score $S_i = F(Z_i, C_i^{\mathrm{tr}})$. Training labels pair ground-truth relations with counterfactual negatives formed by reversing a predicate or replacing an endpoint when the geometry strongly contradicts the label (full rules and thresholds are in the supplement). The resulting score is not a probability of physical validity. Features that combine the predicate with predicate-independent pair measurements belong to the $T_i \times G_i$ term. Neither the source relation score nor predictor identity enters C_i. The same fitted parameters are therefore used for all three predictors on the shared target.

Let $L_K(c)$ be the top-K candidates from the evaluated relation families for context c and Y_c its exact-label ground-truth set. We jointly report

$$\begin{aligned} \mathrm{R@K} &= \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}, \\ \mathrm{Violation@K} &= \frac{1}{|D_K|} \sum_{(c,i)\in D_K} \mathbf{1}[v_i = \mathrm{violated}]. \end{aligned} \tag{5}$$

where D_K contains selected candidates for which the verifier returns $v_i \in \{\mathrm{satisfied}, \mathrm{uncertain}, \mathrm{violated}\}$. Family mapping never relaxes exact predicate matching. Because uncertain candidates enter the denominator, the experiments also report decidable-only and pessimistic violation, uncertainty, and coverage.

Relation-Consistent Compatibility

For each candidate relation r_i, the associated geometry vector G_i contains OBB-derived 3D/XY distances, relative height, projected overlap, and vertical-gap features. Point-level contact evidence is used only by the support/contact verifier, not by the compatibility model. The feature map $\Phi = (\phi_T, \phi_G, \phi_\times)$ separates predicate/family indicators, predicate-independent geometry, and explicit predicate–geometry interactions. The source relation score, rank, predictor identity, and object-class features are excluded.

We first define a family logit and bounded compatibility score

$$\begin{aligned} h_a(\Phi(T_i, G_i)) &= w_a^\top \Phi(T_i, G_i), \\ C_i &= \sigma(h_a(\Phi(T_i, G_i)))\,. \end{aligned} \tag{6}$$

For proximity, τ swaps the ordered endpoints and leaves `close by` unchanged. For vertical order, τ swaps the endpoints and maps `higher than`↔`lower than`. After transformation averaging, the final score satisfies

$$\begin{aligned} C^{\mathrm{tr}}_{\mathrm{close}}(s, o) &= C^{\mathrm{tr}}_{\mathrm{close}}(o, s), \\ C^{\mathrm{tr}}_{\mathrm{higher}}(s, o) &= C^{\mathrm{tr}}_{\mathrm{lower}}(o, s). \end{aligned} \tag{7}$$

Support/contact is evaluated only under the identity transformation because no family-wide endpoint swap preserves every predicate in this family. Its compatibility model is used only in the all-family comparison; the RelCompat3D ranking preserves the source ranking for support/contact.

Training combines transformation-consistent augmentation with a linked-pair ranking loss. The augmentation preserves the predicate when proximity endpoints are swapped, whereas vertical order swaps endpoints and applies the inverse predicate. In addition, for every linked positive–counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logits $\ell = w_a^\top \Phi$ receive a fixed margin penalty:

$$\begin{aligned} \mathcal{L}_a = \mathcal{L}_{\mathrm{BCE}} &+ 0.25\,\mathbb{E}_{\mathcal{P}}\Big[\log\Big(1 + e^{1-(\ell_{i^+} - \ell_{i^-})}\Big)\Big] \\ &+ 10^{-4}\|w_a\|_2^2. \end{aligned} \tag{8}$$

Here $\mathcal{L}_{\mathrm{BCE}}$ is binary cross-entropy over positive and counterfactual examples. Numeric features are standardized and missing values are imputed using statistics estimated from the training split. Each family uses a small linear model fitted exclusively on the training split; optimization details and training-example counts are provided in the supplement.

Finally, we average the learned score over the family-specific transformation set $\mathcal{O}_i$ to obtain the transformation-consistent compatibility:

$$C_i^{\mathrm{tr}} = \frac{1}{|\mathcal{O}_i|} \sum_{(T', G') \in \mathcal{O}_i} \sigma(h_{a_i}(\Phi(T', G')))\,. \tag{9}$$

This transformation averaging makes proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation exact at inference rather than merely encouraged by a penalty. The pairwise objective still distinguishes each linked positive from its counterfactual; transformation averaging does not use the source relation score.

Proposition 1 (Exact transformation consistency) *Let the finite transformation group H_a act jointly on predicate and ordered-pair geometry, and define $C_a^{\mathrm{tr}}(x) = |H_a|^{-1} \sum_{g \in H_a} C_a(gx)$. Then $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$ for every $h \in H_a$. Consequently, Eq. 9 satisfies proximity symmetry and invariance under joint endpoint swap and inverse-predicate transformation for vertical order.*

The supplement gives a proof. Support/contact uses only the identity transformation and is unaffected by this averaging.

Family-Aware Re-ranking

For each candidate, define the within-family ranking score

$$u_i = \begin{cases} Z_i C_i^{\mathrm{tr}}, & a_i \in \mathcal{A}_{\mathrm{tr}}, \\ Z_i, & a_i = \mathrm{support/contact}. \end{cases} \tag{10}$$

For $Z_i, C_i^{\mathrm{tr}} > 0$, $\log u_i = \log Z_i + \log C_i^{\mathrm{tr}}$ in the transformation-supported families; compatibility therefore contributes an additive log-score term without an additional fusion parameter. This rule introduces no fitted fusion parameter.

Let $\pi^Z = (e_1, \ldots, e_n)$ denote the source ranking, a_j the relation family at position j, and π_a the candidates of family a sorted by u_i. At position j, the rule selects the next unused candidate from π_{a_j}, exactly preserving the source family-label sequence. Because support/contact is sorted by Z_i, its selected subsequence and family-specific Recall and Violation are unchanged at every K. For each prefix, the rule also selects the highest-u_i proximity/vertical candidates and maximizes their summed utility subject to the preserved family counts and support/contact subsequence; the supplement gives an exchange proof.

Product, rank-average, RRF, and a matched nonlinear model use the same family-aware ranking procedure. Additional controls test the role of family conditioning, the source relation score, pair identity, geometry, and the defined transformations; full definitions are provided in the experiments and supplement.

Experiments

Experimental Setup

Data and splits. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set relation predictors and enable comparison across predictors on a shared 3DSSG target containing relation families whose consistency can be assessed from reconstructed ordered-pair geometry. All evaluations use the same scope: 157 scans, 548 relation contexts, and 3,972 exact-label ground-truth relations from support/contact, proximity, and vertical-order families. Ground-truth membership is used only when computing Recall@K, never to filter or rank candidate relations.

Open3DSG uses the publicly released model and preprocessing pipeline. Every evaluation context is retained, with unavailable candidate lists treated as empty; coverage and denominator sensitivities are in the supplement.

Baselines and training. We compare the source relation score and RelCompat3D with rank-average, RRF, and a matched multilayer perceptron (matched MLP) using the same family-aware re-ranking procedure. Product (all families) tests the effect of re-ranking support/contact, and a pooled compatibility model tests whether family-specific fitting is necessary. Controls perturb the predicate, pair identity, pair-geometry measurements, endpoint order, source relation score, or distance measurements; exact transformations are in the supplement. The compatibility inputs remain limited to predicate and family indicators T, predicate-independent pair-geometry measurements G, and explicit $T \times G$ interactions. Normalization, imputation, and model parameters

Table 1: **Exact-label Recall (R↑) and verifier-derived Violation (V↓).** All entries are percentages. RelCompat3D, matched MLP, RankAvg, and RRF use the same family-aware ranking procedure; Product (all families) applies compatibility to every relation family.

		$K = 5$		$K = 10$		$K = 20$		$K = 50$		$K = 100$	
Predictor	Ranking rule	R	V	R	V	R	V	R	V	R	V
Open3DSG	Source	3.42	52.05	9.87	32.89	19.89	20.99	40.43	13.87	51.11	12.42
	RelCompat3D	3.73	0.94	11.35	2.33	23.62	3.13	44.18	3.42	56.92	3.24
	Matched MLP	3.70	4.95	11.78	4.97	24.67	4.56	46.70	4.13	59.89	3.71
	RankAvg	3.78	1.09	11.08	2.48	22.41	3.53	43.23	4.58	54.08	5.48
	RRF	3.60	18.99	10.70	16.06	21.63	13.24	42.37	9.48	54.76	7.72
	Product (all families)	9.99	4.05	19.66	4.52	32.85	4.28	46.55	2.65	60.05	3.30
VL-SAT	Source	41.94	0.29	63.22	0.82	80.74	1.42	92.72	2.68	96.35	4.76
	RelCompat3D	42.07	0.15	63.39	0.57	80.82	1.14	92.77	1.97	96.58	2.95
	Matched MLP	42.09	0.15	63.47	0.51	80.92	1.09	92.72	1.89	96.50	2.96
	RankAvg	35.42	0.15	51.76	0.47	69.49	0.99	90.26	1.62	97.00	2.25
	RRF	36.78	0.15	55.49	0.51	74.72	1.04	91.74	1.80	97.10	2.47
	Product (all families)	41.77	0.11	63.32	0.49	80.97	1.10	92.93	2.03	96.88	3.25
SGFN	Source	31.17	2.37	39.75	3.49	49.12	3.22	74.02	3.85	92.35	6.30
	RelCompat3D	31.17	2.37	39.75	3.47	49.14	2.97	74.50	2.63	93.03	3.50
	Matched MLP	31.17	2.37	39.75	3.47	49.19	2.96	74.57	2.58	92.88	3.50
	RankAvg	31.09	2.37	38.80	3.47	46.58	2.95	70.22	2.36	92.80	2.59
	RRF	31.14	2.37	38.92	3.47	47.03	2.97	68.13	2.53	87.74	3.02
	Product (all families)	31.14	2.48	40.61	3.21	51.26	2.74	77.06	2.56	94.18	3.72

use only 1,061 training scans; model design and the applicable relation families were selected on the 117-scan development split. The 157-scan evaluation split is never used for fitting. All cross-predictor statements concern the shared 3DSSG/3RScan target.

A feature-removal analysis refits models exclusively on the training split after excluding either the exact scalar used by the verifier or all related distance or vertical measurements; all conditions are reported in the supplement.

Metrics and uncertainty. Recall@K uses exact labels within each relation context. Violation@K is the fraction of verifier-assessed selected candidates labeled violated. Its denominator includes satisfied, uncertain, and violated candidates, whereas only violated candidates enter the numerator. The decidable-only variant excludes uncertain candidates from the denominator; the pessimistic variant counts them as violations. We report $K \in \{5, 10, 20, 50, 100\}$ and discuss $K = 50$ as an intermediate reported budget. Paired 95% intervals use a scan-cluster bootstrap: all contexts from a sampled scan are resampled jointly, with shared indices across conditions. Within-family results and family-specific metrics within the selected top-K predictions show whether aggregate changes are driven by relation-family composition.

Recall–Violation Results

Table 1 and Figure 2 show that the family-aware RelCompat3D re-ranking yields a better $K = 50$ point-estimate Recall–Violation trade-off for all three predictors. On Open3DSG and SGFN, paired scan-cluster intervals are strictly positive for Recall and negative for Violation. On VL-SAT, whose Source Recall is already high, the Recall interval contains zero while the Violation interval remains strictly negative, supporting lower verifier-derived Violation without a detectable Recall loss. Open3DSG exposes the failure most strongly: the Source baseline ranks geometrically inconsistent relation candidates highly at small K, whereas RelCompat3D improves Recall and reduces Violation without discarding candidates. Across $K = 10$–50, no RelCompat3D Recall point estimate is lower than its Source baseline, and the Violation point estimate decreases for every predictor. SGFN changes only marginally at $K = 10$, while Open3DSG exhibits the largest gains throughout this range.

No single comparator dominates across all predictors and reported K values. Under the same ranking procedure, the matched MLP raises Open3DSG Recall more than the RelCompat3D product at $K = 50$ (46.70% versus 44.18%) but also has higher Violation (4.13% versus 3.42%); the two are nearly tied on VL-SAT and SGFN. The comparisons expose predictor-dependent trade-offs: on VL-SAT and SGFN, the RelCompat3D product avoids the larger low-budget Recall losses observed with rank fusion. On Open3DSG, the matched MLP raises Recall at the cost of higher Violation. Product (all families) reaches higher aggregate Recall and can lower aggregate Violation, but changes support/contact selections and can mask a support/contact regression. The pooled family-conditioning ablation is reported in the supplement. The matched MLP, rank-average, and RRF use the same family-aware re-ranking procedure as RelCompat3D; they therefore preserve the relation-family sequence and keep support/contact candidates in source order. Product (all families) does not impose this restriction.

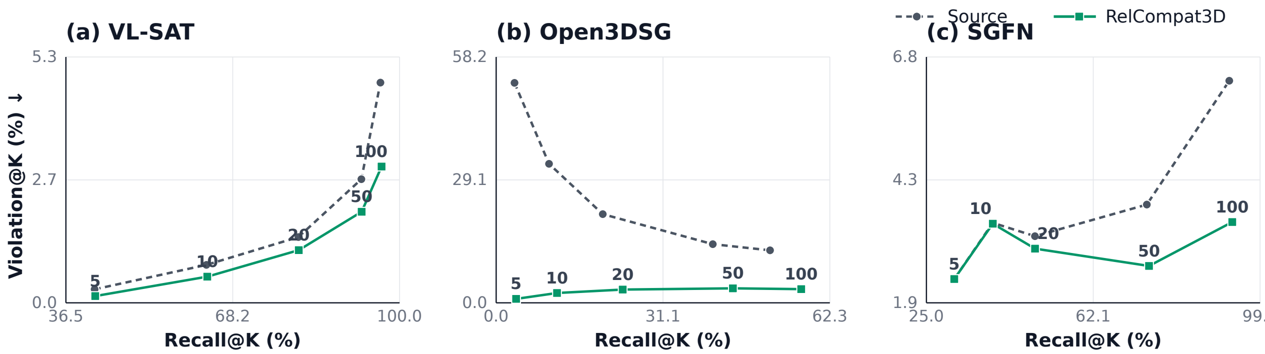

Figure 2: **Recall@K–Violation@K trajectories.** Each line connects $K \in \{5, 10, 20, 50, 100\}$ for one predictor and ranking rule. Rightward and downward movement is preferred. Labels mark K on the RelCompat3D curve; Source baseline points follow the same sequence. Axis ranges differ by predictor.

Ablations and Controls

Table 2 tests changes to the predicate, pair, geometry, endpoint order, and score inputs under the same family-aware ranking procedure as the main method.

Table 2: **Ablations and counterfactual controls.** All entries are percentages; R is Recall (↑) and V is Violation (↓). Every condition uses the same candidates and family-aware ranking procedure as Table 1.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	**RelCompat3D**	44.18	3.42	56.92	3.24
	Wrong predicate	42.65	22.00	53.47	20.98
	Wrong pair	38.52	8.48	49.32	8.35
	Shuffled geometry	38.44	12.93	48.69	12.43
	Label-fixed swap	42.67	22.00	53.73	20.98
	Distance only	51.16	8.24	63.22	9.55
	Compatibility only	42.07	3.42	56.77	3.24
VL-SAT	**RelCompat3D**	92.77	1.97	96.58	2.95
	Wrong predicate	90.43	4.98	94.81	7.96
	Wrong pair	90.53	2.61	94.96	4.73
	Shuffled geometry	89.17	3.06	94.11	5.60
	Label-fixed swap	90.43	4.98	94.81	7.96
	Distance only	81.90	5.34	89.80	8.09
	Compatibility only	67.65	1.61	84.09	2.03
SGFN	**RelCompat3D**	74.50	2.63	93.03	3.50
	Wrong predicate	71.55	9.43	89.98	12.99
	Wrong pair	71.58	3.99	87.99	6.65
	Shuffled geometry	70.62	4.74	86.78	8.36
	Label-fixed swap	71.55	9.43	90.03	12.99
	Distance only	63.19	10.00	84.06	12.66
	Compatibility only	52.79	2.32	70.72	2.30

On the development split, linked positives outrank their counterfactuals in 99.23% of pairs, and correct vertical predicates outrank their inverses in 97.40% of cases. Transformation averaging makes the maximum compatibility difference under a proximity endpoint swap exactly zero. It also makes the maximum difference under a joint endpoint swap and inverse-predicate transformation exactly zero for vertical order, on development and across all evaluation candidates. These results verify the implemented transformation behavior and support the intended counterfactual ordering.

The product without transformation averaging has nearly identical aggregate Recall, while averaging removes the applicable transformation-consistency errors exactly. The input decomposition reports T-only, predicate-independent G-only, additive, and interaction conditions in the supplement.

Wrong-pair and shuffled geometry reduce Recall and raise Violation. The result therefore depends on associating each prediction with the geometry of its corresponding ordered pair rather than on scene-level geometric statistics alone. Wrong predicates and label-fixed swaps sharply increase Violation. Here, label-fixed swap exchanges the subject and object while retaining the original predicate label. For binary vertical predicates, this endpoint swap creates the same sign contradiction as an inverse-predicate transformation, while proximity is swap-invariant. Distance alone raises Open3DSG Recall but more than doubles Violation relative to RelCompat3D and is worse on both metrics for VL-SAT and SGFN. Compatibility-only ranking omits the source relation score; it retains low Violation but loses substantial Recall, especially on VL-SAT and SGFN. The all-family product comparison is reported in the supplement. Together, these controls support the interpretation that the observed trade-off depends on predicate-conditioned measurements from the corresponding ordered pair, combined with—not substituted for—the source relation score.

The feature-removal analysis examines direct reuse of verifier measurements. Removing the exact scalar used by the verifier preserves nearly the full $K = 50$ result. Removing all related distance or vertical measurements attenuates the effect: Open3DSG still improves both metrics, while VL-SAT and SGFN retain Recall but their Violation approaches the Source baseline. The compatibility model therefore does not depend solely on a single verifier scalar, although related geometric measurements remain important.

Qualitative and Family Analysis

Figure 3 shows two relations whose consistency can be assessed from reconstructed ordered-pair geometry and one

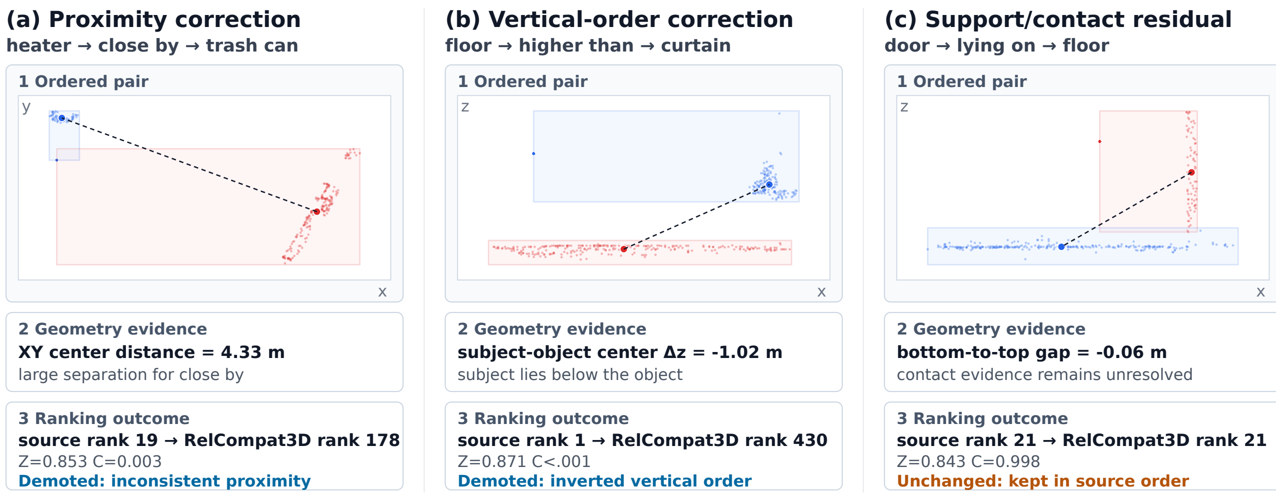

Figure 3: **Pair–evidence–outcome analysis.** Each column links an ordered-pair point-cloud view to its measured geometric evidence and ranking outcome. The first two violations are demoted; the contact-heavy residual is kept in source order. Subjects use circular points and solid boxes; objects use square points and dashed boxes, in addition to red/blue color.

residual support/contact case that remains in source order. The method moves the heater–trash-can proximity edge from rank 19 to 178 because the pair centers are 4.33 m apart, and moves `floor higher than curtain` from rank 1 to 430 because the floor center is 1.02 m below the curtain. It leaves the residual `door lying on floor` support candidate at rank 21; Product (all families) would promote it to rank 10, illustrating why the current method keeps support/contact in source order.

Family-wise intervals show that the aggregate change is not uniform across relation families. Proximity and vertical order both contribute to Recall changes, while vertical order accounts for most of the Violation reduction; support/contact selection and metrics are unchanged by construction. Support/contact errors therefore remain uncorrected. Full within-family results and family-specific metrics within the selected top-K predictions are provided in the supplement.

Discussion and Limitations

The results indicate that predicate–geometry compatibility is most useful when relation consistency can be assessed from reconstructed ordered-pair geometry and the source candidate set contains recoverable alternatives. The compatibility layer is evaluated across three predictors on one shared 3DSSG target; these results characterize cross-predictor behavior, not cross-dataset generalization. A separate ReplicaSSG/FROSS stress test shows that the observed gains do not transfer uniformly under ontology and reconstructed-geometry shifts (supplement).

The aggregate effect is not uniform across relation families. By keeping support/contact candidates in source order, family-aware re-ranking leaves that family's Recall and Violation unchanged, but it also leaves its errors uncorrected. Family-specific metrics within the selected top-K predictions are reported separately. RelCompat3D assumes known object instances and is limited to relation families whose consistency can be assessed from reconstructed ordered-pair geometry. Functional and reference-frame-dependent predicates and complete open-vocabulary graph generation remain outside scope.

Violation@K is verifier-derived, and some geometry primitives overlap with the constructed training target. Wrong-predicate, wrong-pair, and transformation controls test whether the compatibility score responds to predicate–geometry correspondence. Feature-removal experiments show that the results do not depend solely on the exact scalar used by the verifier. Attenuation after removing all related distance or vertical measurements nevertheless does not replace an independent physical-validity reference. Independent validity labels, richer contact evidence, and evaluation on datasets with different geometry and relation ontologies would be needed to support broader validity and generalization claims.

Conclusion

RelCompat3D separates predicate–geometry compatibility from the source relation score and enforces the defined proximity and vertical endpoint/predicate transformations. Across three predictors on a shared 3DSSG target, family-aware re-ranking yields lower verifier-derived Violation point estimates without reducing Recall point estimates at $K = 10$–50, while support/contact candidates remain in source order.

References

Armeni, I.; He, Z.-Y.; Gwak, J.; Zamir, A. R.; Fischer, M.; Malik, J.; and Savarese, S. 2019. 3D Scene Graph: A Structure for Unified Semantics, 3D Space, and Camera. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 5664–5673.

Chen, L.; Wang, X.; Lu, J.; Lin, S.; Wang, C.; and He, G. 2024. CLIP-Driven Open-Vocabulary 3D Scene Graph Generation via Cross-Modality Contrastive Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 27863–27873.

Chen, X.; Hu, J.; Yang, Y.; Zhang, X.; Wang, T.; Bao, H.; Zhang, G.; and Cui, Z. 2026. Beyond Isolated Objects: Relationship-Aware Open-Vocabulary Scene Understanding via 3D Scene Graph Analysis. arXiv:2607.05348.

Feng, C.; Yin, W.; Hu, X.; Zhao, L.; Zhang, D.; and He, T. 2026. GEODE: Geometry-Guided Discrete Diffusion for Open-Vocabulary 3D Scene Graph Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings*, 7143–7153.

Feng, M.; Hou, H.; Zhang, L.; Wu, Z.; Guo, Y.; and Mian, A. 2023. 3D Spatial Multimodal Knowledge Accumulation for Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9182–9191.

Gu, Q.; Kuwajerwala, A.; Morin, S.; Jatavallabhula, K. M.; Sen, B.; Agarwal, A.; Rivera, C.; Paul, W.; Ellis, K.; Chellappa, R.; Gan, C.; de Melo, C. M.; Tenenbaum, J. B.; Torralba, A.; Shkurti, F.; and Paull, L. 2024. ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 5021–5028.

Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 1321–1330.

Heo, K.; Kim, G.; Kim, S.; and Cho, M. 2025. Object-Centric Representation Learning for Enhanced 3D Semantic Scene Graph Prediction. In *Advances in Neural Information Processing Systems*.

Hou, H.-Y.; Lee, C.-Y.; Sonogashira, M.; and Kawanishi, Y. 2025. FROSS: Faster-Than-Real-Time Online 3D Semantic Scene Graph Generation from RGB-D Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 28818–28827.

Huang, I.; Bao, Y.; Truong, K.; Zhou, H.; Schmid, C.; Guibas, L.; and Fathi, A. 2025. FirePlace: Geometric Refinements of LLM Common Sense Reasoning for 3D Object Placement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13466–13476.

Hughes, N.; Chang, Y.; and Carlone, L. 2022. Hydra: A Real-Time Spatial Perception System for 3D Scene Graph Construction and Optimization. In *Robotics: Science and Systems (RSS)*.

Koch, S.; Hermosilla, P.; Vaskevicius, N.; Colosi, M.; and Ropinski, T. 2024a. SGRec3D: Self-Supervised 3D Scene Graph Learning via Object-Level Scene Reconstruction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 3404–3414.

Koch, S.; Vaskevicius, N.; Colosi, M.; Hermosilla, P.; and Ropinski, T. 2024b. Open3DSG: Open-Vocabulary 3D Scene Graphs from Point Clouds with Queryable Objects and Open-Set Relationships. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14183–14193.

Liu, Y.; Mei, H.; Zhan, D.; Zhao, J.; Zhou, D.; Dong, B.; and Yang, X. 2026. View-on-Graph: Zero-Shot 3D Visual Grounding via Vision-Language Reasoning on Scene Graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7386–7394.

Ma, Y.; Liu, H.; Guo, Y.; Gevers, T.; and Oswald, M. R. 2026. Edge-Centric Relational Reasoning for 3D Scene Graph Prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7847–7855.

Madhavaram, V.; Sengar, V.; De, A.; and Sharma, C. 2026. VIZOR: Viewpoint-Invariant Zero-Shot Scene Graph Generation for 3D Scene Reasoning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 8584–8595.

Maggio, D.; Chang, Y.; Hughes, N.; Trang, M.; Griffith, J. D.; Dougherty, C.; Cristofalo, E.; Schmid, L.; and Carlone, L. 2024. Clio: Real-Time Task-Driven Open-Set 3D Scene Graphs. *IEEE Robotics and Automation Letters*, 9(10): 8921–8928.

Neau, M.; Baloch, S.; Suchan, J.; Falomir, Z.; and Bhatt, M. 2026. Visual Commonsense Driven Knowledge Refinements for Scene Graph Generation. *arXiv preprint arXiv:2606.06369*.

Nguyen, M. A.; Tran, Q. H.; Le, B. N.; Pham, T. K.; and Guang, S. Y. 2026. RelWitness: Open-Vocabulary 3D Scene Graph Generation with Visual-Geometric Relation Witnesses. arXiv:2605.20823.

Peng, S.; Genova, K.; Jiang, C. M.; Tagliasacchi, A.; Pollefeys, M.; and Funkhouser, T. 2023. OpenScene: 3D Scene Understanding With Open Vocabularies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 815–824.

Sarkar, S. D.; Miksik, O.; Pollefeys, M.; Barath, D.; and Armeni, I. 2023. SGAligner: 3D Scene Alignment with Scene Graphs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 21927–21937.

Saxena, P.; and Chiun, J. 2025. ZING-3D: Zero-shot Incremental 3D Scene Graphs via Vision-Language Models. arXiv:2510.21069.

Shao, Y.; Zhai, W.; Yang, Y.; Luo, H.; Cao, Y.; and Zha, Z.-J. 2025. GREAT: Geometry-Intention Collaborative Inference for Open-Vocabulary 3D Object Affordance Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17326–17336.

Sun, J.; Wang, C.; Liu, Z.; Tian, J.; Yang, M.; Wang, Y.; and Gao, S. 2026. Not All Relations Rotate Alike: Transformation-Aware Decoupling for Viewpoint-Robust 3D Scene Graph Generation. arXiv:2606.27412.

Takmaz, A.; Fedele, E.; Sumner, R. W.; Pollefeys, M.; Tombari, F.; and Engelmann, F. 2023. OpenMask3D: Open-Vocabulary 3D Instance Segmentation. In *Advances in Neural Information Processing Systems*, volume 36, 68367–68390.

Wald, J.; Dhamo, H.; Navab, N.; and Tombari, F. 2020. Learning 3D Semantic Scene Graphs From 3D Indoor Reconstructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3961–3970.

Wang, Z.; Cheng, B.; Zhao, L.; Xu, D.; Tang, Y.; and Sheng, L. 2023. VL-SAT: Visual-Linguistic Semantics Assisted Training for 3D Semantic Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 21560–21569.

Wang, Z.; Su, Y.; Li, C.; Wang, D.; Huang, Y.; Li, X.; and Zhao, B. 2025. Open-Vocabulary Octree-Graph for 3D Scene Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 7037–7047.

Werby, A.; Huang, C.; Büchner, M.; Valada, A.; and Burgard, W. 2024. Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation. In *Robotics: Science and Systems (RSS)*.

Wu, S.-C.; Wald, J.; Tateno, K.; Navab, N.; and Tombari, F. 2021. SceneGraphFusion: Incremental 3D Scene Graph Prediction From RGB-D Sequences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 7515–7525.

Xie, Y.; Pagani, A.; and Stricker, D. 2024. SG-PGM: Partial Graph Matching Network with Semantic Geometric Fusion for 3D Scene Graph Alignment and Its Downstream Tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 28401–28411.

Yang, Y.; Castillo, M.; Rosenhahn, B.; and Yang, M. Y. 2026. PUF: Plug-and-Play Uncertainty-Aware Fusion for Online 3D Scene Graph Generation. Accepted at ECCV 2026, arXiv:2607.07170.

Yeo, Q. X.; Li, Y.; and Lee, G. H. 2025. Statistical Confidence Rescoring for Robust 3D Scene Graph Generation from Multi-View Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 24999–25008.

Yu, F.; Deng, Q.; Tang, S.; Li, Y.; and Cheng, L. 2025. Open-World 3D Scene Graph Generation for Retrieval-Augmented Reasoning. Accepted by AAAI 2026, arXiv:2511.05894.

Zhang, C.; Delitzas, A.; Wang, F.; Zhang, R.; Ji, X.; Pollefeys, M.; and Engelmann, F. 2025. Open-Vocabulary Functional 3D Scene Graphs for Real-World Indoor Spaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 19401–19413.

Zhang, C.; Yu, J.; Song, Y.; and Cai, W. 2021. Exploiting Edge-Oriented Reasoning for 3D Point-Based Scene Graph Analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9705–9715.